

9 Power Supply Recommendations

The TPS63900 device is designed to operate with input supplies from 1.8 V to 5.5 V. The input supply must be stable and free of noise to achieve the full performance of the device. If the input supply is located more than a few centimeters away from the device, additional bulk capacitance can be required. The input capacitance shown in the application schematics in this data sheet is sufficient for typical applications.

10 Layout

10.1 Layout Guidelines

PCB layout is an important part of any switching power supply design. A poor layout can cause unstable operation, load regulation problems, increased ripple and noise, and EMI issues.

The following PCB layout design guidelines are recommended:

- Place the input and output capacitors close to the device.
- Minimize the area of the input loop, and use short, wide traces on the top layer to connect the input capacitor to the VIN and GND pins.
- Minimize the area of the output loop, and use short, wide traces on the top layer to connect the output capacitor to the VOUT and GND pins.
- The location of the inductor on the PCB is less important than the location of the input and output capacitors. Place the inductor after the input and output capacitors have been placed close to the device. You can route the traces to the inductor on an inner layer if necessary.

10.2 Layout Example

Figure 10-1 shows an example of a PCB layout that follows the recommendations of the previous section.

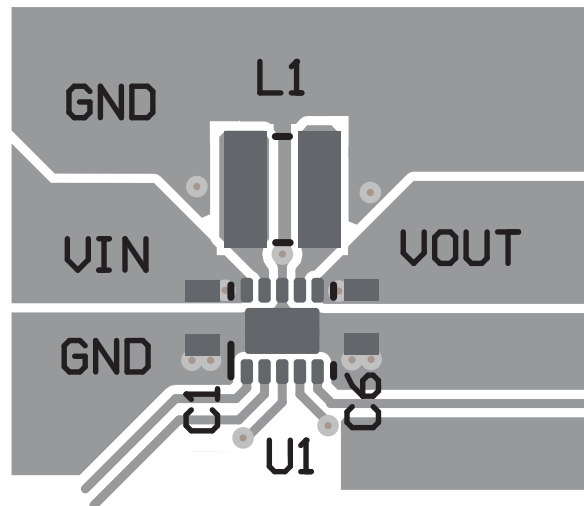


Figure 10-1. PCB Layout Example